

User Guide: Enterprise AI Decision Intelligence Platform

This guide explains how business users and technical users should use the application.

1. What The App Does

The application helps teams review AI/ML decision models, understand their business impact, and inspect the technical evidence behind each run.

It is organized around four model workflows:

- TensorFlow Next Best Action Model
- TensorFlow Churn Propensity Model
- Bayesian Media Mix Optimization
- Causal Incrementality Model

Each workflow moves from model inputs to recommendation, impact forecast, decision evidence, activation output, and governance evidence.

2. Recommended Presentation Flow

Use this sequence for an end-to-end walkthrough:

- Start at AI Decision Portfolio.
- Open AI Impact Summary to show portfolio-level health.
- Open the Model Decision Workbench.
- Walk through model inputs, recommendation, impact forecast, and decision evidence.
- Open Simulation Flow to change setup and run the AI/ML pipeline one stage at a time.
- Open Architecture Reference to explain the technical runtime and evidence path.

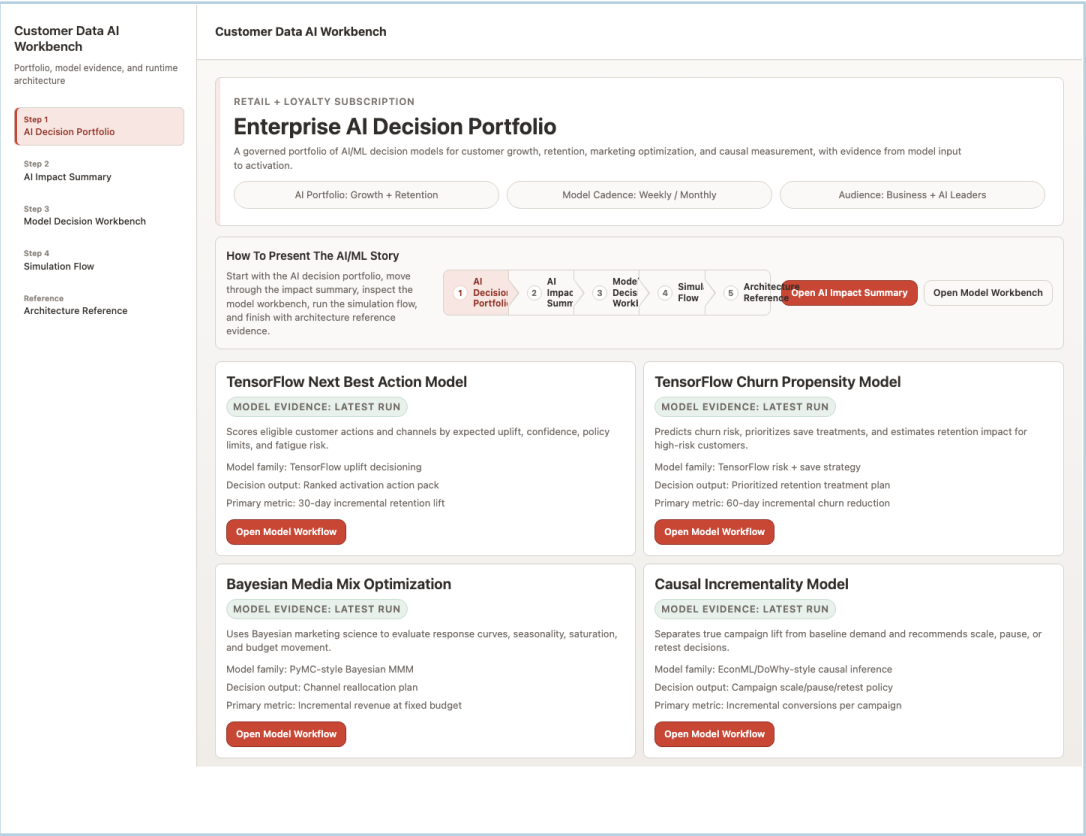
3. Visual Walkthrough

The screenshots below show the recommended flow a user should follow during a business or technical walkthrough.

Flow Step 1: Start With The AI Decision Portfolio

Use this page to introduce the model portfolio and explain the four AI/ML decision workflows.

AI Decision Portfolio



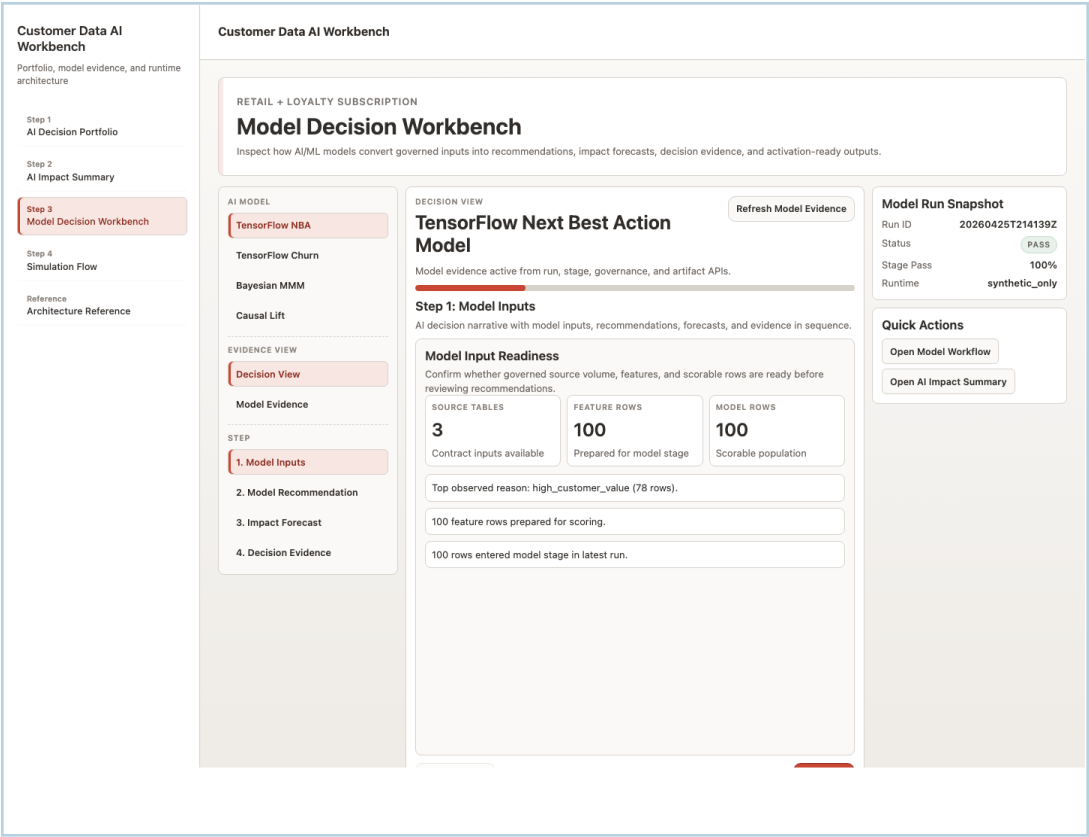
What to point out:

- Four governed AI/ML decision models.
- Each model has a clear business purpose and primary metric.
- The recommended presentation sequence starts with customer-level AI decisioning.

Flow Step 2: Open The Model Decision Workbench

Open a model workflow and start with Model Inputs. This is where users see the data readiness and scorable population before looking at recommendations.

Model Inputs



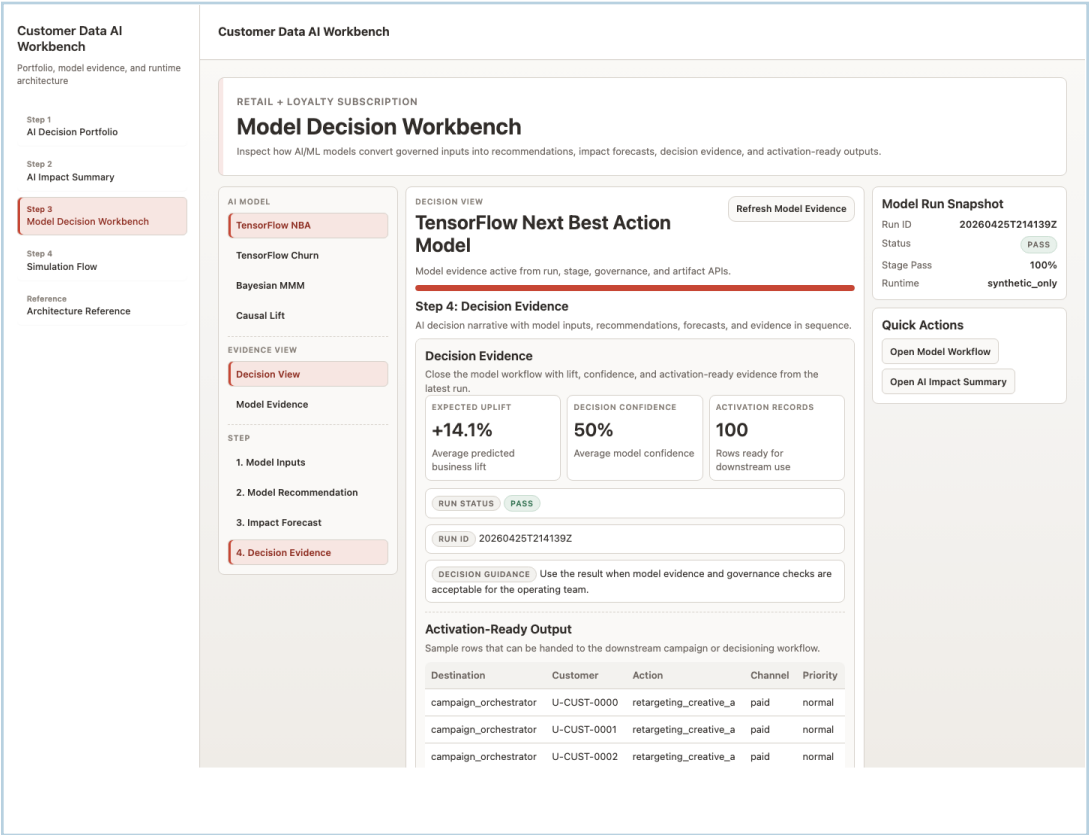
What to point out:

- The left navigation keeps model, evidence view, and step selection visible.
- The center panel explains one decision step at a time.
- The right-side snapshot confirms run ID, status, stage pass rate, and runtime.

Flow Step 3: Review Decision Evidence

Move to Decision Evidence to close the business workflow with uplift, confidence, activation records, and run status.

Decision Evidence



What to point out:

- Expected uplift summarizes predicted business value.
- Decision confidence shows the average model confidence signal.
- Activation records show that outputs are ready for downstream decisioning.

Flow Step 4: Review AI Impact Summary

Open AI Impact Summary when you need a portfolio-level executive view.

AI Impact Summary

Customer Data AI Workbench

Portfolio, model evidence, and runtime architecture

Step 1
AI Decision Portfolio

Step 2
AI Impact Summary

Step 3
Model Decision Workbench

Step 4
Simulation Flow

Reference
Architecture Reference

Customer Data AI Workbench

RETAIL + LOYALTY SUBSCRIPTION

AI Impact Summary

One-page executive view of AI model impact, readiness, and decision evidence across the model portfolio. Each card summarizes the model family, recommendation direction, and measurable business signal.
Live model evidence active: sourced from /api/run-history and /api/runs across all four AI/ML use cases.

Refresh Model Evidence

Print / Save PDF

AI Model Portfolio Snapshot

MODELS REVIEWED

4

MODELS HEALTHY

4/4

AVERAGE PIPELINE PASS

100%

LAST REFRESH (UTC)

2026-04-28T02:44:52.621Z

TensorFlow Next Best Action Model

Run 20260425T214139Z

Decision: Approve the model-ranked action pack for highest-value eligible customers.
Model Flow: Inputs -> Recommendation -> Impact -> Evidence

Expected Uplift

+14.1%

Decision Confidence

50%

Run Status

Pass

Pipeline health: 100% stages passed

Runtime: synthetic_only | Scenario Preset: default

Review Model Workflow

TensorFlow Churn Propensity Model

Run 20260423T032819Z

Decision: Prioritize model-recommended save treatments for high-risk subscribers.
Model Flow: Inputs -> Recommendation -> Impact -> Evidence

Average Churn Risk

33.7%

Expected Save Lift

+7.9%

Run Status

Pass

Pipeline health: 100% stages passed

Runtime: unknown | Scenario Preset: default

Review Model Workflow

Bayesian Media Mix Optimization

Run 20260423T032828Z

Decision: Approve Bayesian budget shifts toward channels with the strongest marginal return.
Model Flow: Inputs -> Recommendation -> Impact -> Evidence

Forecast Error (MAPE)

90.1%

Channels Planned

60

Run Status

Pass

Pipeline health: 100% stages passed

Runtime: unknown | Scenario Preset: default

Review Model Workflow

Causal Incrementality Model

Run 20260423T220328Z

Decision: Scale campaigns with measured causal lift and pause weak performers.
Model Flow: Inputs -> Recommendation -> Impact -> Evidence

Average True Lift

+0.5%

Average iROAS

0.98x

Run Status

Pass

Pipeline health: 100% stages passed

Runtime: unknown | Scenario Preset: default

Review Model Workflow

What to point out:

- Models reviewed and models healthy.
- Average pipeline pass rate.
- Model-level decisions, KPIs, and latest run status.

Flow Step 5: Run The Simulation Flow

Open Simulation Flow to change the AI/ML use case, runtime profile, scenario preset, seed, or failure injection, then run the pipeline stage by stage.

Simulation Flow

Customer Data AI Workbench

Portfolio, model evidence, and runtime architecture

Step 1
AI Decision Portfolio

Step 2
AI Impact Summary

Step 3
Model Decision Workbench

Step 4
Simulation Flow

Reference
Architecture Reference

Customer Data AI Workbench

ENTERPRISE AI/ML PLATFORM ARCHITECTURE

Simulation Flow

Change the model setup, run the AI/ML pipeline one stage at a time, and watch runtime evidence update as the flow progresses.

CHANGE & SIMULATE FLOW

Choose a model setup, run the pipeline one stage at a time, and watch the evidence update.

UC-NBA-RET-001. Simulation has not started. Model execution available. Recommended path: Integrated AI Runtime.

1
Change setup
Use case, runtime, scenario

2
Simulate
Run Next Stage
Reset

3
Watch evidence
Open stage walkthrough

PRESENTATION STEPS

01
Simulation flow

02
AI/ML Architecture Walkthrough

03
Enterprise AI hardening

04
Model evidence

AI/ML USE CASE

UC-NBA-RET-001 - Next Best Action for Retention

ARCHITECTURE PROFILE

Standalone AI Runtime

Enterprise Integration Profile exposes optional lineage, feature-store, identity, orchestration, and access-control readiness while the default runtime stays standalone.

MODEL EXECUTION

☒ Selected AI/ML use case only Seed 101 Data Runtime default

Scenario Preset none

☐ Inject DQ fail ☐ Inject schema fail ☐ Inject backend unavailable

Run Integrated AI Runtime Run Standalone AI Runtime Run Next Stage Pause Simulation Resume Simulation

Reset Simulation

Model execution available. Recommended path: Integrated AI Runtime.

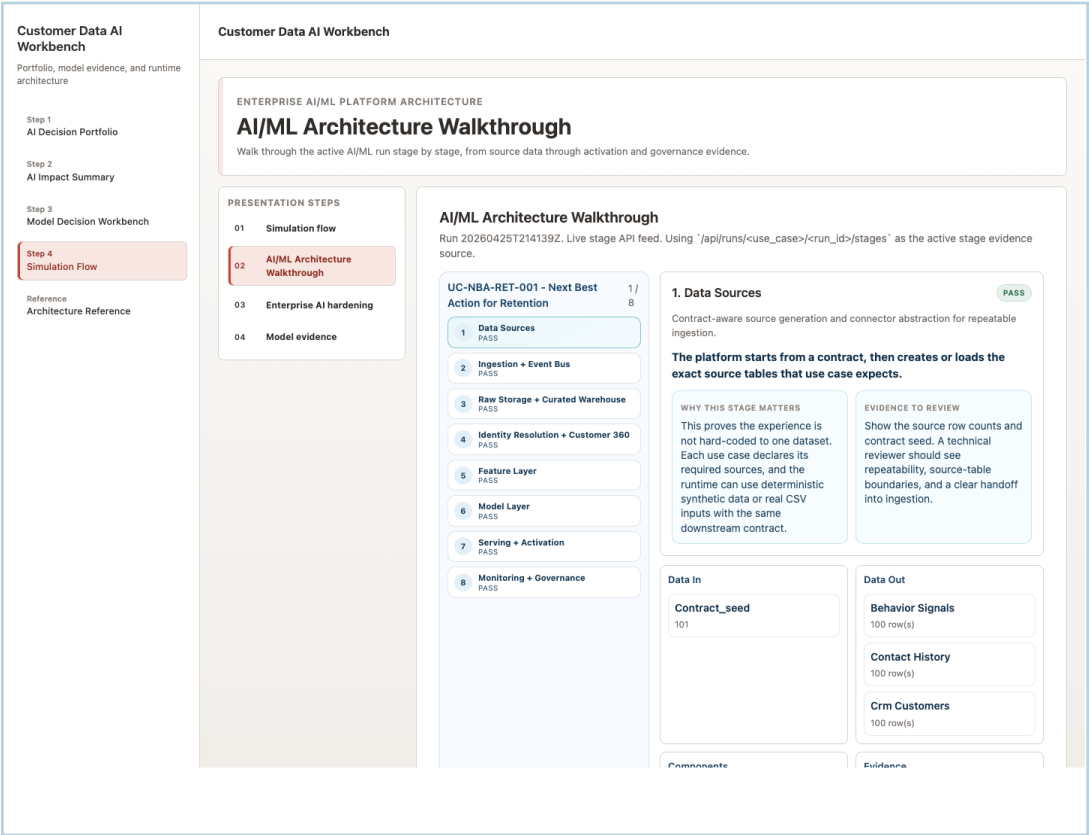
What to point out:

- AI/ML Use Case is a dropdown so users can switch models.
- Model Execution controls expose runtime, scenario, seed, and failure simulation.
- Run Next Stage advances the walkthrough and updates runtime evidence.

Flow Step 6: Walk The AI/ML Architecture

Use AI/ML Architecture Walkthrough to explain the pipeline from Data Sources to Monitoring + Governance.

AI/ML Architecture Walkthrough



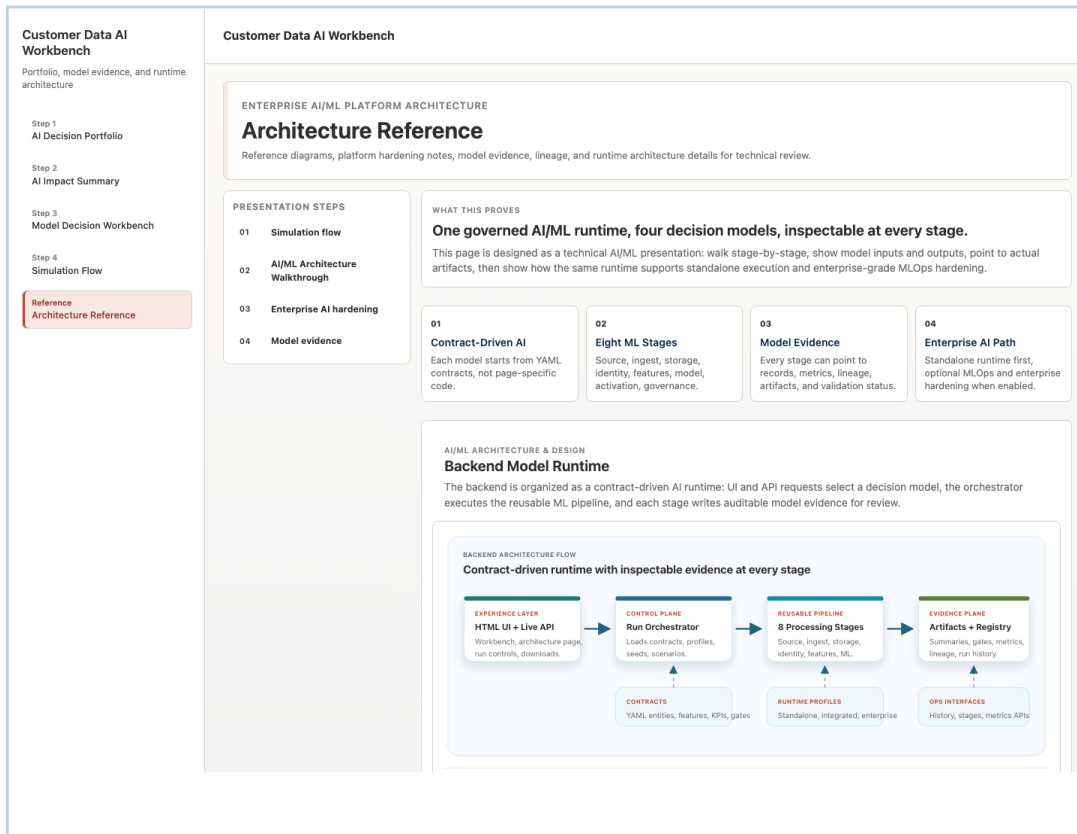
What to point out:

- Each stage has a runtime responsibility.
- Each stage produces artifacts or metrics.
- Technical users can inspect inputs, outputs, components, and evidence.

Flow Step 7: Review Architecture Reference And Evidence

Use Architecture Reference and Model Evidence to review the reference diagrams, architecture flow, run history, and portfolio health.

Architecture Reference And Evidence



What to point out:

- Architecture Reference separates static reference from action-oriented simulation.
- Model Evidence shows run artifacts and execution evidence.
- Model Run History compares recent runs.
- AI Portfolio Health summarizes latest run status across use cases.

Flow Step 8: Switch To Technical Model Evidence

In the Model Decision Workbench, switch from Decision View to Model Evidence for a deeper technical review.

Technical Model Evidence

The screenshot displays the 'Customer Data AI Workbench' interface. On the left, a sidebar lists steps: Step 1 (AI Decision Portfolio), Step 2 (AI Impact Summary), Step 3 (Model Decision Workbench - highlighted), Step 4 (Simulation Flow), and a Reference Architecture Reference. The main content area is titled 'Model Decision Workbench' under the 'RETAIL + LOYALTY SUBSCRIPTION' header. It shows the 'TensorFlow Next Best Action Model' with a 'Refresh Model Evidence' button. Below this, 'Step 3: Responsible AI Evidence' is detailed, including a 'Governance + Evidence' section with 'not_applicable' data quality status, '0/0' blockers/warnings, and 'candidate' registry stage. A 'Governance Meta' table shows 'Run Status' as 'pass', 'Run ID' as '20260425T214139Z', and 'Monitoring Report' as 'download'. A 'Validation Gates' table shows 'uplift_validation_pass' and 'calibration_pass' both as 'PASS'. On the right, a 'Model Run Snapshot' shows 'Run ID' '20260425T214139Z', 'Status' 'PASS', 'Stage Pass' '100%', and 'Runtime' 'synthetic_only'. Below this, a 'Model Evidence Snapshot' shows 'Components' '8/8 pass', 'Artifacts' '17/28 present', 'Data Quality' 'not_applicable (0/0)', and 'Model Backend' 'tensorflow'. 'Quick Actions' include 'Open Model Workflow' and 'Open AI Impact Summary'.

What to point out:

- Contract controls show use case, runtime, scenario, and model constraints.
- Technical evidence connects components, APIs, stage outputs, and artifacts.
- The same workflow can be explained to business users or technical users without changing apps.

4. Starting The App

From the repository root:

```
make standalone
```

Open:

```
http://127.0.0.1:8080/ui/experience/index.html
```

If the environment is already prepared:

```
make standalone-fast
```

For live UI/API mode:

```
make ui-live
```

5. Main Pages

Page | URL | Primary Audience

AI Decision Portfolio | /ui/experience/business-home.html | Business and executive users

AI Impact Summary | /ui/experience/business-exec-summary.html | Executives and business owners

Model Decision Workbench | /ui/experience/workbench.html | Business users and technical reviewers

Simulation Flow | /ui/experience/index.html#simulate | Architects, ML engineers, and platform teams

Architecture Reference | /ui/experience/index.html#overview | Architects, ML engineers, and platform teams

6. Business User Guide

Business users should focus on the decision story and model impact.

Step 1: Open AI Decision Portfolio

Use the portfolio page to understand which models are available and what each model is designed to decide.

Look for:

- Model name
- Decision output

- Primary business metric
- Recommended presentation sequence

Step 2: Open A Model Workflow

Click Open Model Workflow for one of the models.

Recommended first workflow:

TensorFlow Next Best Action Model

Step 3: Walk The Business Steps

In the Model Decision Workbench, use the left-side step navigation:

- Model Inputs
- Model Recommendation
- Impact Forecast
- Decision Evidence

Use these questions while presenting:

- What customer, campaign, or spend signals entered the model?
- What does the model recommend?
- What impact does the model forecast?
- What evidence supports the decision?

Step 4: Review The AI Impact Summary

Open AI Impact Summary to see portfolio-level model health.

Look for:

- Models reviewed
- Models healthy

- Average pipeline pass rate
- Latest run status
- Model-specific business KPIs

Step 5: Share Or Print

Use Print / Save PDF on the AI Impact Summary page when you need an executive-ready output.

7. Technical User Guide

Technical users should focus on architecture, run evidence, artifacts, and governance.

Step 1: Open Simulation Flow

Use Simulation Flow to change the model setup and run the end-to-end platform one stage at a time.

The core architecture is:

Data Sources

- > Ingestion + Event Bus
- > Raw Storage + Curated Warehouse
- > Identity Resolution + Customer 360
- > Feature Layer
- > Model Layer
- > Serving + Activation
- > Monitoring + Governance

Step 2: Walk The AI/ML Architecture

Use AI/ML Architecture Walkthrough to walk stage by stage.

For each stage, review:

- What the stage does
- Data in
- Data out
- Components
- Evidence artifacts

Step 3: Use Model Evidence View

Open Model Decision Workbench and select Model Evidence.

Use this view to inspect:

- Contract controls
- Stage inputs and outputs
- Model backend
- Model metrics
- Activation payloads
- Validation gates
- Artifact paths

Step 4: Inspect Architecture Reference And Model Evidence

Open Architecture Reference for static diagrams, then open Model Evidence for run artifacts and history.

Review:

- Architecture Reference
- Model Evidence
- Model Run History
- AI Portfolio Health

Step 5: Run A New Model Workflow

Use Model Execution controls on Simulation Flow.

Common options:

- Select an AI/ML use case.
- Choose Standalone AI Runtime or Integrated AI Runtime.
- Choose runtime mode.
- Select a scenario preset if needed.
- Click Run Standalone AI Runtime or Run Integrated AI Runtime.

Step 6: Simulate Stage Execution

Use the simulation controls to explain the pipeline gradually:

- Run Next Stage
- Pause Simulation
- Resume Simulation
- Reset Simulation

This is useful when presenting the architecture to technical audiences.

8. Runtime Profiles

Profile | Use When

Standalone AI Runtime | You want the app to run locally without external services.

Integrated AI Runtime | You want to show optional Kafka, Postgres, or MinIO-style infrastructure mirroring.

Enterprise Integration Profile | You want to show readiness for MLflow, Feast, Splink, Airflow, Keycloak, monitoring, and metrics.

The app can run standalone without deployment dependencies. Enterprise integrations are optional.

9. Common Tasks

Run all model workflows

```
python3 -m pipelines.cli run-stack-all
```

Run one workflow

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001
```

Run synthetic-only mode

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 --runtime-mode  
synthetic_only
```

View recent runs

```
python3 -m pipelines.cli list-runs
```

Inspect run stages

```
python3 -m pipelines.cli show-stages --use-case UC-NBA-RET-001 --run-id <run_id>
```

Inspect model readiness

```
python3 -m pipelines.cli model-readiness --use-case UC-NBA-RET-001 --run-id <run_id>
```

10. Troubleshooting

If the app is not loading:

- Confirm the server is running.
- Confirm the URL uses the right port.
- Run make standalone-fast if dependencies and artifacts already exist.
- Run make smoke-standalone to validate the app.

If model evidence is missing:

- Run at least one workflow.
- Refresh the page.
- Check artifacts/<use_case_id>/<run_id>/summary.json.

If integrated runtime is unavailable:

- Use Standalone AI Runtime.
- Confirm optional infrastructure is running before using Integrated AI Runtime.

11. What To Show In A Meeting

For business stakeholders:

- AI Decision Portfolio
- One Model Workflow
- AI Impact Summary

For technical stakeholders:

- AI/ML Platform Architecture
- Eight ML stages
- Model Evidence View
- Model Run History
- Runtime Evidence

For platform leaders:

- Business value from model workflows
- Evidence-backed architecture
- Standalone runtime
- Enterprise integration path